CITS5508 Practice Exam — Chapter 3 — ANSWER KEY

Question 1.

(a) *(2 marks)* TP = 40, TN = 80, FP = 20, FN = 10.

(b) *(6 marks)* - **Accuracy** = (TP + TN) / total = (40 + 80) / 150 = 120/150 = **0.80 (80%)**. - **Precision** = TP / (TP + FP) = 40 / (40 + 20) = 40/60 = **0.667 (66.7%)**. - **Recall** = TP / (TP + FN) = 40 / (40 + 10) = 40/50 = **0.80 (80%)**. - $\mathbf{F_1}$ = 2·(P·R)/(P+R) = 2·(0.667·0.80)/(0.667+0.80) = 2·0.5333/1.4667 = **0.727**. *(1½ marks each.)*

(c) *(4 marks)* On a **skewed/imbalanced** dataset, a model that simply predicts the **majority (negative) class** can score high accuracy without detecting any positives — so accuracy hides poor positive-class performance. Report metrics derived from the **confusion matrix**: **precision**, **recall**, $\mathbf{F_1}$, and look at the **PR curve** (positive class is rare). *(Here recall = 80% is fine, but precision = 67% shows real weakness that accuracy masks.)*

Question 2.

(a) *(4 marks)* The classifier scores each instance via a **decision function** and predicts positive if the score exceeds a **threshold**. **Raising** the threshold makes fewer positive predictions → **precision tends to increase** (the remaining positives are more confidently correct) and **recall decreases** (more true positives are missed). You cannot maximise both at once — improving one generally worsens the other.

(b) *(4 marks)* 1. Kid-safe video filter → **high precision** (better to wrongly reject some safe videos than to let one unsafe video through; a false positive "safe" label is costly). 2. Tumour screening → **high recall** (must catch nearly all true tumours; missing one (false negative) is dangerous, while false alarms can be re-checked). *(2 marks each: correct choice + justification.)*

(c) *(4 marks)* As the threshold rises one instance at a time, the **recall** denominator (total actual positives) is fixed and TP can only drop or stay → recall is **monotonically non-increasing** (smooth). **Precision** = TP/(TP+FP); removing a positive prediction can remove either a TP or an FP, which can move the TP/(TP+FP) ratio **up or down** depending on which it was — so precision is **bumpy / not strictly monotonic**.

Question 3.

(a) *(4 marks)* The ROC curve plots, over **all thresholds**: - **y-axis: True Positive Rate (TPR = recall) = TP/(TP+FN)** — fraction of positives caught. - **x-axis: False Positive Rate (FPR) = FP/(FP+TN) = 1 − specificity** — fraction of negatives wrongly flagged positive. Each point corresponds to one threshold.

(b) *(4 marks)* **AUC** = area under the ROC curve = the probability that the classifier ranks a random **positive** above a random **negative** (a threshold-independent measure of ranking quality). **Perfect classifier AUC = 1.0**; **random classifier AUC = 0.5** (the diagonal). A good classifier's curve **bows toward the top-left corner** (high TPR at low FPR); the further from the diagonal, the better.

(c) *(4 marks)* With rare positives there are **many easy negatives** (large TN), which keeps FPR low even when there are many false positives — so the ROC/AUC can look **deceptively good**. The **PR curve** uses precision = TP/(TP+FP) and recall = TP/(TP+FN), **neither of which involves TN**, so it directly exposes how many of the positive predictions are actually wrong — more informative when the positive class is rare or false positives are costly.

Question 4.

(a) *(2 marks)* "free offer" → Money 0, Free 1, Win 0, Offer 1, Meeting 0, Report 0 → **q = [0, 1, 0, 1, 0, 0]**.

(b) *(5 marks)* Squared Euclidean distances (then √): - to #1 [2,1,1,0,0,0]: $(0-2)^2+(1-1)^2+(0-1)^2+($ = 4+0+1+1 = **6** → √6 ≈ **2.449** - to #2 [1,1,0,1,0,0]: $(0-1)^2+0+0+0+0+0$ = **1** → √1 = **1.000** - to #3 [0,0,0,0,1,1]: 0+1+0+1+1+1 = **4** → √4 = **2.000** - to #4 [0,0,0,1,1,1]: 0+1+0+0+1+1 = **3** → √3 ≈ **1.732**

(c) *(3 marks)* - **k = 1:** nearest is **#2** (distance 1.000) → predict **True (spam)**. - **k = 3:** three nearest are #2 = 1.000 (True), #4 = 1.732 (False), #3 = 2.000 (False). Votes = 1 True, 2 False → predict **False (not spam)**. *(Note the prediction flips from spam at k=1 to not-spam at k=3.)*

(d) *(2 marks)* Feature scaling matters because (any two): k-NN uses **distances**, so a feature with a **large numeric range dominates** the distance and swamps small-range features; without scaling, distances (and therefore neighbours) reflect units rather than relevance. The bias/variance hyperparameter is **k** (the number of neighbours): small k = low bias/high variance (overfit), large k = high bias/low variance (underfit).

Question 5.

(a) *(5 marks)* - **Multiclass:** exactly **one** label per instance, chosen from **> 2 mutually exclusive** classes. *Example:* handwritten-digit recognition — features = pixel intensities; response ∈ {0,…,9}. - **Multilabel: multiple** (independent) binary labels per instance simultaneously. *Example:* tagging which people appear in a photo — features = image pixels; response = {Alice?, Bob?, Charlie?} (a set of binary tags). *(2 marks distinction, 1½ per correctly described example.)*

(b) *(4 marks)* - **OvR:** train **N** binary classifiers (one class vs all the rest); predict the class whose classifier gives the highest score. - **OvO:** train one classifier per **pair** of classes → **N·(N−1)/2** classifiers; predict the class winning the most duels. - OvO is preferred for **SVMs** because SVM training **scales poorly with dataset size** (≈ $O(m^2$–$m^3)$); OvO trains each classifier on only the **two relevant classes' data** (small subsets), which is faster than a few classifiers on the full large set.

(c) *(3 marks)*

```
y_scores = forest.predict_proba(X)[:, 1]   # probability of the positive class
```

`predict_proba` returns estimated **class probabilities**; column 1 is the estimated probability of the **positive class**, used as the ranking score for `precision_recall_curve`.